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PAPER

Mechanism design and experiment verification of a mole-inspired robot burrowing with incisors

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Abstract

With the growing demand for underground resources, traditional drilling equipment faces significant limitations in soil environments. In recent years, bionic burrowing robots have attracted increasing research attention for their potential advantages in miniaturization, adaptability, and low energy consumption, although their development is still in the early exploratory stage. This study presents a mole-inspired robot designed based on the remarkable burrowing capability of the naked mole-rat (Heterocephalus glaber), which uses its incisors to break soil and limbs to propel itself forward. The incisor mechanism of the robot achieves a single-degree-of-freedom (DOF) occlusion via a gear drive and linkage transmission system. To analyze the relationship between the incisor tip force and the servo output torque, a mechanical model based on the principle of virtual work and virtual displacement is established, and its accuracy is validated through physical experiments. The leg mechanism employs a Chebyshev-parallelogram composite linkage configuration to achieve single-DOF forward–backward leg motion. To ensure optimal kinematic performance, the leg kinematics are analyzed, and the leg link lengths are optimized through foot-end trajectory planning. Finally, a prototype was developed and tested in soils with varying moisture contents. The experimental results verify the proposed design methodology and mechanical model, confirming the feasibility and effectiveness of the mole-inspired incisor-limb coordination strategy for autonomous burrowing.

1. Introduction

As the demand for underground resource development continues to increase, technologies for exploring subsurface environments have become a critical research frontier. The extreme complexity of these environments–characterized by soil collapsibility, intricate geological stratifications, and diverse rock structures–poses significant challenges to conventional detection and sampling methods. Although techniques such as drilling, seismic wave analysis, and geophysical imaging provide partial insights into subsurface conditions, they are often limited by high costs and low efficiency [1]. Furthermore, extreme environments such as volcanic regions, deep-sea formations, and planetary soils introduce additional difficulties related to equipment transport and energy supply [2]. Zhang *et al* summarized the recent advancements and trends in planetary soil sampling technologies [3]. Traditional drilling systems are typically limited by large size, high power consumption, and poor portability. In addition, restrictions on payload capacity and energy supply impose stringent constraints on the size, weight, mobility, power consumption, and efficiency of drilling equipment [4–6].

With advancements in bionics, researchers have drawn inspiration from natural organisms to design compact, lightweight, and energy-efficient robots capable of operating in subsurface environments [7, 8]. These bio-inspired robots demonstrate superior mobility and flexibility [9, 10]. Moreover, their excavation depth is not limited by their size, and they

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eliminate the need for extensive surface infrastructure. As a result, bio-inspired burrowing robots have emerged as a key research focus in underground excavation technology. The biological inspiration for such robots often originates from animals that forage or live underground, such as earthworms, moles, and clams. By mimicking their structural configurations and locomotion mechanisms, the burrowing capabilities of robots can be significantly enhanced. Recent studies have reviewed the current state and development trends of bio-inspired burrowing robots for both terrestrial and planetary soil excavation [11, 12].

The mole is a small mammal renowned for its outstanding burrowing capability, often referred to as a 'living excavator'. They can burrow soil equivalent to their body volume in a remarkably short period. Compared with other animals, they exhibit outstanding burrowing capabilities and remarkable adaptability to the complex conditions of underground environments. In particular, moles are capable of burrowing in various soil types, including loose, hard, and moist soils, thereby maintaining high burrowing performance across diverse conditions. Moles exhibit variations in their primary burrowing methods depending on their habitats. Based on these strategies, moles can be categorized into two types: either burrowing with forelimbs (forelimb-scratching mole) [13] or burrowing with incisors (incisor-chiseling mole) [14]. The former one primarily employs their powerful forelimbs to penetrate and displace soft or moist soils through large-amplitude burrowing strokes, whereas the latter one utilizes their robust incisors and reinforced cranial structures to cut and fracture compact or granular soils. Adapted to arid or coarse-grained environments, naked-mole rats exhibit efficiency in excavating harder substrates than the former one, highlighting their particular significance for research and engineering applications.

Currently, relevant research teams have developed multiple mole-inspired robots, which can be categorized into three main types based on their movement mechanisms: hammering-compaction robots, forelimb-burrowing robots, and hybrid incisor-forelimb robots. Hammering-compaction robots mimic the burrowing actions of forelimb-type moles, precisely disrupting soil structure and achieving efficient compaction during excavation. This mechanism enables efficient penetration in dense or cohesive soils and provides stable self-anchoring during drilling, making it particularly suitable for deep subsurface sampling, planetary regolith exploration, and anchoring tasks on low-gravity celestial bodies. However, their performance depends strongly on soil mechanical impedance and moisture conditions, and the high-intensity vibration leads to considerable energy consumption and potential structural wear. Representative examples include the mole robot developed by Gromov *et al* in 1997 [15], the PLUTO robot developed by Lutz Richter *et al* in 2002 [16], and the MMUM robot developed by Carol Stoker *et al* in 2007 [17]. Forelimb-burrowing robots mimic the movement patterns of mole forelimbs, where scapulae drive the forelimbs to extend forward and retract backward, effectively transferring excavated soil from the front to the rear. They are particularly suitable for soft or medium-density soils, where controlled excavation and minimal vibration are required-for example, in planetary surface sampling and shallow subsurface exploration. However, their overall efficiency and maintenance convenience remain limited by single-operation capacity and flexibility. Notable examples include the earth-burrowing rescue robot proposed by Kobayashi *et al* in 2011 [18], the USAR mobile burrowing robot presented by Richardson *et al* in 2011 [19], and the mole-inspired robot developed by Wei *et al* in 2024 [20]. Hybrid incisor-forelimb robots integrate the crushing capability of incisors with the propulsion and burrowing functions of forelimbs, enabling coordinated crushing-burrowing operations that greatly enhance excavation efficiency and optimize energy utilization. These hybrid systems apply to mixed or stratified regolith layers, where both high cutting strength and flexible propulsion are required. However, the coupled actuation systems increase mechanical complexity, overall mass, and power consumption, posing challenges to system reliability under extraterrestrial conditions. A prominent example is the Mole-Bot system proposed by Hyun Myung *et al* in 2020 [21].

While existing mole-inspired robots primarily rely on forelimb-based mechanisms or hybrid systems, these designs often face limitations in efficiency, particularly when dealing with compact or cohesive soils. In contrast, the incisor-based burrowing approach offers distinct advantages. The incisor design concentrates cutting forces at the tooth-soil interface, allowing for more efficient soil penetration compared with the dispersed forces exerted by forelimb-based systems. This results in improved performance, especially in challenging soils, where the incisors can directly cut through the soil with less energy loss. Moreover, the incisor-based approach generates significantly lower vibrations during operation. Unlike forelimb-driven robots, which rely on large, forceful limb strokes to displace soil, the incisor mechanism minimizes reactive forces, enhancing both the stability of the robot and the surrounding environment. These features make the incisor-inspired robot particularly suitable for applications that require high precision and adaptability to varying soil types, such as underground exploration, planetary regolith excavation, and disaster response.

Overall, the unique biological structures and locomotion mechanisms of moles provide valuable

inspiration for the design of burrowing robots. Most existing mole-inspired robots primarily focus on the forelimbs of forelimb moles as the basis for their design, with relatively limited research dedicated to incisor-chiseling moles. The mechanical advantage of the masticatory muscles of the incisor-chiseling mole is significantly higher than that of the forelimb-scratching mole [22], and naked mole (Heterocephalus glaber) incisors exhibit excellent mechanical properties, making them excellent objects for bionic mechanical system design [23]. Inspired by Heterocephalus glaber, this study designs a mole-inspired robot burrowing with incisors that can achieve autonomous burrowing. The robot utilizes the function of incisors to break soil so that it can adapt to soil layers of different hardnesses, and combines the structural design and locomotion principles of legged robots to achieve coordinated excavation and propulsion during the burrowing process. The remainder of this paper is organized as follows: section 2 introduces the biological mechanisms of the naked-mole rat and the design and analysis of the primary bionic mechanisms of the robot. Section 3 introduces the development of the prototype and the related experimental testing. Section 4 concludes the study and outlines future work.

2. Design and analysis of bionic mechanisms

2.1. Bioinspired design of the robot

Naked-mole rats are a group of rodents adapted to subterranean life, primarily inhabiting arid regions. In such environments, the lack of moisture leads to densely compacted and hard soils, which in turn have endowed these moles with specialized burrowing abilities. Compared with forelimb-scratching moles, naked-mole rats are capable of breaking harder soils and exhibit greater adaptability. Their burrowing behavior primarily relies on the biting action of their incisors to break soil, while their limbs serve key functions in stabilizing their bodies, enabling forward propulsion, and expelling soil from the burrow. During the burrowing process, naked-mole rats first anchor themselves using their hind limbs and vertebral column. Then, they utilize their incisors to break the soil and employ agile head movements to deposit the soil in front of their bodies. Once the soil accumulates to a certain extent, it is pushed downward by the forelimbs and kicked backward by the hind limbs, effectively clearing it from the tunnel and preventing blockages, as shown in figures 1(a) and (b).

The outstanding burrowing capability of naked-mole rats is closely related to their specialized structures. Their body is primarily composed of the head, limbs, spine, and tail, which coordinate to enable efficient burrowing. The skull of moles features a broad oral cavity, expanded zygomatic arches, and large temporal fossae that accommodate well-developed muscle groups, including the superficial masseter, deep masseter, and temporal muscles (figure 1(d1)) [22, 24], thereby generating substantial biting force. During the biting process, moles open their mouths while contracting their lips laterally, retracting the lower jaw, and closing the nasal valves to prevent soil from entering [26]. Despite the limited biting range of their incisors, their flexible head movements allow them to dig a hole significantly larger than their body size. In addition, the upper and lower incisors play different roles: the upper incisors provide anchorage, while the lower incisors perform the biting action. Notably, the upper incisors are not completely immobile, but move a smaller distance than the lower incisors [27]. Currently, the hind limbs of moles remain relatively understudied. They consist mainly of the bones of the thigh, calf, and pasterns [28], as shown in figure 1(e1).

Inspired by the outstanding burrowing capability and unique biological structure of naked-mole rats, we have developed a mole-inspired robot that burrows with incisors. The overall structure of the robot is designed biomimetically based on three main components: a head, four legs, and a buttock, as shown in figure 1(c). The robot possesses a total of six degrees-of-freedom (DOFs): two in the head and four in the legs. The buttock, primarily used for sealing and housing the control system, has no active DOFs. The design of the primary burrowing mechanism is described in detail next.

Inspired by the unique head structure of the naked-mole rat, we developed a biomimetically designed incisor mechanism. Based on kinematic analysis of mole head motion, key joint locations were identified and mapped to corresponding points on the skull structure of the mole [27], as shown in figure 1(d1). These points informed the structural design of the incisor mechanism. The core design concept is to replicate the opening and closing motion of incisors of the mole to achieve soil breaking. To this end, the complex head structure of the mole was simplified into a drive unit and two linkage assemblies. For compactness, transmission efficiency, and system stability, a gear-based transmission system was selected. Servo-driven gears actuate the linkages, enabling the incisor mechanism to open and close. This design ensures high precision and torque output while meeting spatial constraints. Prior analysis determined the occlusion mode as a double incisor occlusion, and the optimal opening angle was set to 75° [29]. A 3D model of the incisor mechanism is shown in figure 1(d2). To facilitate soil removal from between the incisors and increase the occlusion area, an additional DOF was added to the head of the robot, allowing it to rotate. The overall head design is scaled up to the actual mole head ratio of 2:1, which is largely dictated by engineering constraints. This ratio reflects

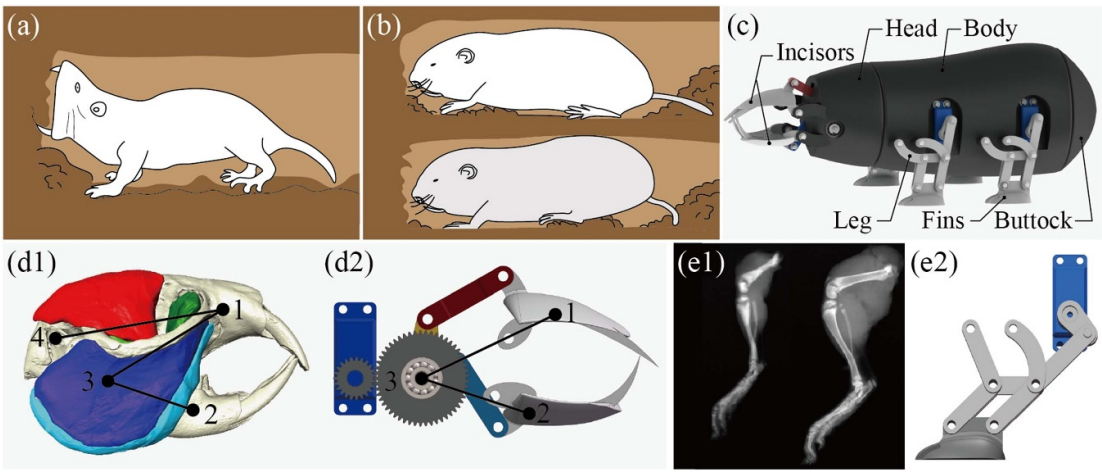

Figure 1. Bioinspired design of the robot. (a) Soil breaking of incisor. (b) Removal of soil from limbs. (c) 3D model of the whole robot. (d1) Model of key points of skeletal muscles in the mole head. Reproduced from [24]. CC BY 4.0 (d2) 3D model of the incisor mechanism. (e1) Scan of the leg bones of the mole. Reproduced from [25], with permission from Springer Nature. (e2) 3D model of leg mechanism.

Table 1. Head size comparison between the robot and animal mole [30].

Head size	Robot (mm)	Mole (mm)
Length	99	48.39 ± 5.02
Height	69	36.59 ± 4.13
Width	70	39.13 ± 2.13

the size compatibility of the actuator and drive train and is designed to achieve the required bite force in a limited space. This configuration ensures sufficient output force and penetration depth while maintaining low inertia. Table 1 presents a specific size comparison.

The robotic leg mechanism is biomimetically inspired by the skeletal structure of moles. Based on biomimetic principles, a single-DOF leg mechanism was constructed by integrating a Chebyshev linkage with a parallelogram mechanism. The leg mechanism is driven to lift the leg to take a step by the full-cycle rotation of the crank, which avoids the impact and energy loss caused by the traditional articulated actuator commutation requirement, thus improving the driving efficiency and leg motion performance [31]. Furthermore, by controlling the full-cycle rotation of the crank either in the forward or reverse direction, the leg mechanism can achieve both forward and backward movement. The 3D model of the leg mechanism is shown in figure 1(e2).

2.2. Mechanical analysis of the incisor mechanism

Mechanical analysis of the incisor mechanism is essential for understanding its force transmission characteristics. A static force analysis is conducted to establish the relationship between the output force of the incisor tip and the output torque of the servo, thereby validating the rationality of the design. The principle of virtual work-virtual displacement is employed to construct a mechanical model of the incisor-torque relationships without changing the equilibrium state of the system (the force output of the incisor tip is equivalent to the force received by the incisor tip). At a given moment, under the influence of T_1, F_1, and F_2, the linkages undergo virtual displacements, placing the mechanism in static equilibrium, as shown in figure 2. Here, T represents the torque, F_1 and F_2 denote the forces acting on the incisor tip at that moment, and δ represents the virtual displacement of each linkage. The numerical values of the geometric parameters of the incisor used in the model are summarized in table 2.

Firstly, a virtual displacement equation is constructed for the upper incisor mechanism. Let θ_1 represent the deflection angle of crank AB relative to the x-axis, and β_1 denote the angle formed between the line connecting the two fixed supports (AD) and the x-axis. Since link BC undergoes planar motion, the virtual displacements of points B and C (δr_B and δr_C) have equal projections along link BC. By introducing auxiliary lines AD and BD, the relationship between δr_B and δr_C can be determined through geometric analysis, as expressed by the following equation:

$$\delta r_B \cos\varphi_2 = \delta r_C \cos\varphi_3 \tag{1}$$

where $\varphi_2 = \angle ABD + \angle DBC - \pi/2$ and $\varphi_3 = \pi/2 - \angle BCD$, which can be obtained from $\triangle ABD$ and $\triangle DBC$ by the cosine theorem ($2ab\cos\alpha = a^2 + b^2 - c^2$); α denotes the included angle between linkage elements; a, b, and c represent the corresponding side lengths of the geometric triangle; and $\triangle$ indicates the geometric triangle used for calculation. The specific parameter list is provided in table 3. In addition, $\angle BAD = \theta_1 - \beta_1$.

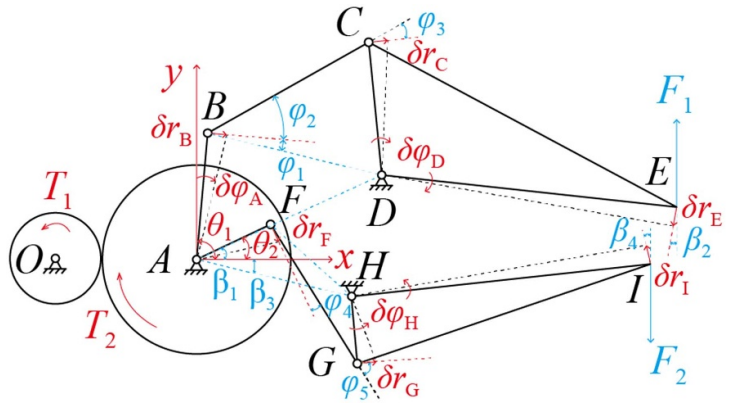

Figure 2. Schematic diagram of the mechanical analysis of the incisor mechanism.

Table 2. Geometric parameters of the incisor.

Parameter	l_{AB}	l_{AD}	l_{BC}	l_{CD}	l_{ED}	l_{AF}	l_{AH}	l_{FG}
Value	15.5	27.46	25.0	17.88	43.7	11.0	21.59	22.0
Parameter	l_{GH}	l_{IH}	$\angle BAF$	β_1	β_3	Z_1	Z_2	—
Value	8.98	43.23	60	24.44	13.39	23	53	

Units: length in millimeter (mm); angle in degree (°); Z_1 and Z_2 are teeth number.

Table 3. Geometric relationships and parameters of upper incisor.

α	a	b	c	$\triangle$
$\angle ABD$	l_{AB}	l_{BD}	l_{AD}	$\triangle ABD$
$\angle DBC$	l_{BD}	l_{BC}	l_{CD}	$\triangle DBC$
$\angle BCD$	l_{BC}	l_{CD}	l_{BD}	$\triangle DBC$
$\angle BAD$	l_{AB}	l_{AD}	l_{BD}	$\triangle ABD$

Table 4. Geometric relationships and parameters of lower incisor.

α	a	b	c	$\triangle$
$\angle AFH$	l_{AF}	l_{FH}	l_{AH}	$\triangle AFH$
$\angle HFG$	l_{FG}	l_{FH}	l_{HG}	$\triangle HFG$
$\angle FGH$	l_{FG}	l_{GH}	l_{FH}	$\triangle HFG$
$\angle FAH$	l_{AF}	l_{AH}	l_{FH}	$\triangle AFH$

By incorporating the following virtual displacement equations, the relationship between the δr_E and δr_A can be found:

$$\begin{cases} \delta r_B = \delta\varphi_A l_{AB} \\ \delta r_C = \delta\varphi_D l_{DC} \\ \delta r_E = \delta\varphi_D l_{ED}. \end{cases} \tag{2}$$

Then, a virtual displacement equation is constructed for the lower incisor mechanism. Let θ_2 represent the deflection angle of crank AF relative to the x-axis, and β_2 denote the angle formed between the line connecting the two fixed supports and the x-axis. Since link FG undergoes planar motion, the virtual displacements of points F and G (δr_F and δr_G) have equal projections along link FG. By introducing auxiliary lines AH and FH, the relationship between δr_F and δr_G can be determined through geometric analysis, as expressed by the following equation:

$$\delta r_F \cos\varphi_4 = \delta r_G \cos\varphi_5 \tag{3}$$

where, $\varphi_4 = \angle AFH - \angle HFG - \pi/2$, $\varphi_5 = \pi/2 - \angle FGH$, and similarly can be obtained from $\triangle AFH$ and $\triangle HFG$ by the cosine theorem, with the specific parameter list shown in table 4. In addition, $\theta_2 = \theta_1 - \angle BAF$, $\angle FAH = \theta_2 + \beta_3$.

By incorporating the following virtual displacement equations, the relationship between δr_I and δr_A can be found:

$$\begin{cases} \delta r_F = \delta\varphi_A l_{AF} \\ \delta r_G = \delta\varphi_H l_{HG} \\ \delta r_I = \delta\varphi_H l_{IH}. \end{cases} \tag{4}$$

Finally, based on the principle of virtual work and the gear torque transmission formula ($T_2 Z_1 = T_1 Z_2$), the relationship between the forces at the incisor tip (F_1 and F_2) and the output torque of the servo (T_1) can be derived, as shown in equation (5).

$$\begin{cases} F_1 = \dfrac{T_2 \delta\varphi_A}{\delta r_E \cos\beta_2} \\ F_2 = \dfrac{T_2 \delta\varphi_A}{\delta r_I \cos\beta_4}. \end{cases} \tag{5}$$

In order to understand the mechanical characteristics of the incisor mechanism more specifically, numerical calculations and analyses were performed

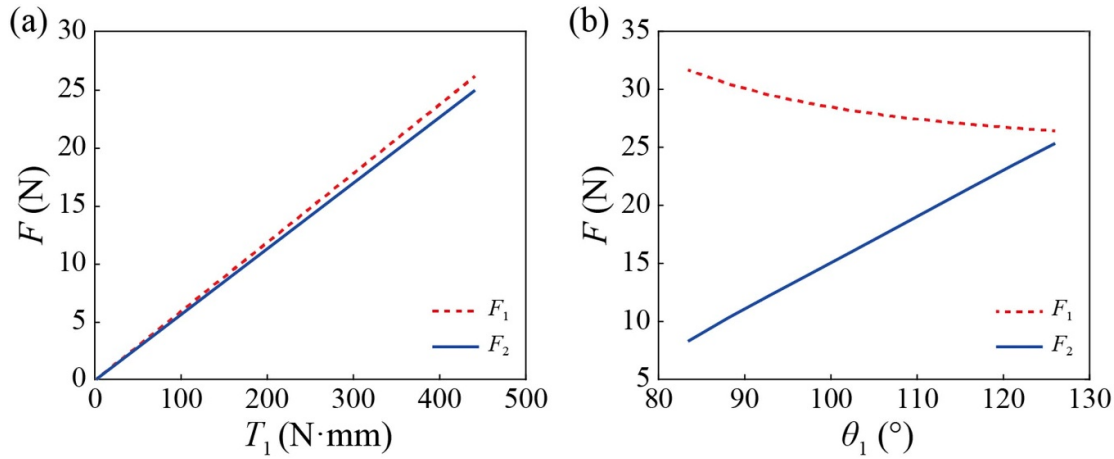

Figure 3. Schematic force curve at the incisor tip. (a) At a fixed θ_1, F_1 and F_2 change with T_1. (b) At a fixed T_1, F_1 and F_2 change with θ_1.

with the aid of MATLAB. The variation of F_1 and F_2 with respect to T_1 was analyzed under the condition $\theta_1 = 0.694\pi$ (i.e. the maximum opening angle of the incisor), as shown in figure 3(a). The results indicate that both F_1 and F_2 increase with T_1 when θ_1 is fixed. Notably, F_1 changes more significantly and its force is larger. When the servo delivers maximum torque, F_1 reaches up to 26 N and F_2 up to 25 N, which are sufficient forces for effective soil breaking.

In addition, the variation of F_1 and F_2 with θ_1 when $T_1 = 441$ N·mm (i.e. the maximum output torque of the servo), as shown in figure 3(b). The results show that as θ_1 increases, F_1 gradually decreases while F_2 increases. This trend can be attributed to the changing orientation of the incisors: as the upper incisors open, the angle between their virtual displacement and the horizontal axis decreases, leading to a reduction in F_1. In contrast, the corresponding angle for the lower incisors increases, resulting in an increase in F_2. The reason for this phenomenon is that in the working process of incisors, the upper incisors serve primarily for anchoring and the lower incisors for occlusion, which is similar to the role played by the incisors of actual mole.

2.3. Kinematic analysis and optimization of the leg mechanism

To ensure stable robot locomotion, foot-end trajectory planning is essential. First, a kinematic analysis of the leg mechanism was performed. Based on this, the foot-end trajectory was planned and used as the optimization objective in a mathematical model [32, 33]. Next, the *fmincon* function in MATLAB was employed to determine the optimal leg mechanism parameters. Finally, motion simulation was conducted using ADAMS to verify the feasibility of the optimized leg mechanism dimensions.

The kinematic analysis of the leg mechanism was conducted, as shown in figure 4. Based on the motion geometry of the mechanism, when the position of the crank is known, the coordinates of the end point D depend on the lengths of the various links (denoted as $l_1 \cdots l_{10}$, and h) and the crank angle θ_1. The coordinates of point A are set as $(l_0, 0)$. To simplify calculations, auxiliary lines l_{AG} and l_{BG} are constructed, and thus:

$$\begin{cases} x_B = x_A - l_1 \cos\theta_1 \\ y_B = l_1 \sin\theta_1 \\ l_{AG} = \sqrt{l_{10}^2 + h^2} \\ \varphi_1 = \arctan\left(\dfrac{h}{l_{10}}\right) \\ l_{BG} = \sqrt{l_1^2 + l_{AG}^2 - 2l_1 l_{AG}\cos(\pi - \theta_1 - \varphi_1)}. \end{cases} \tag{6}$$

Using the auxiliary α_1 and α_2 in $\triangle AGB$ and $\triangle BGC$, θ_2 can be determined as:

$$\begin{cases} \alpha_1 = \arccos\left(\dfrac{l_{BG}^2 + l_{AG}^2 - l_1^2}{2l_{BG}l_{AG}}\right) \\ \alpha_2 = \arccos\left(\dfrac{l_{BG}^2 + l_7^2 - l_2^2}{2l_{BG}l_7}\right) \\ \theta_2 = \alpha_1 + \alpha_2 - \varphi_1. \end{cases} \tag{7}$$

The coordinates of point C can be determined using equation (7):

$$\begin{cases} x_C = x_A + l_{10} - l_7 \cos\theta_2 \\ y_C = l_7 \sin\theta_2 + h. \end{cases} \tag{8}$$

Since points B, C, and D are collinear, the coordinates of point D can be determined using the section formula for proportional division:

$$\begin{cases} p = \dfrac{l_3 + l_2}{l_2} \\ x_D = x_C - p(x_B - x_C) \\ y_{D=y_C - p(y_B - y_C)}. \end{cases} \tag{9}$$

The foot-end trajectory of the robot is composed of the support phase and the swing phase.

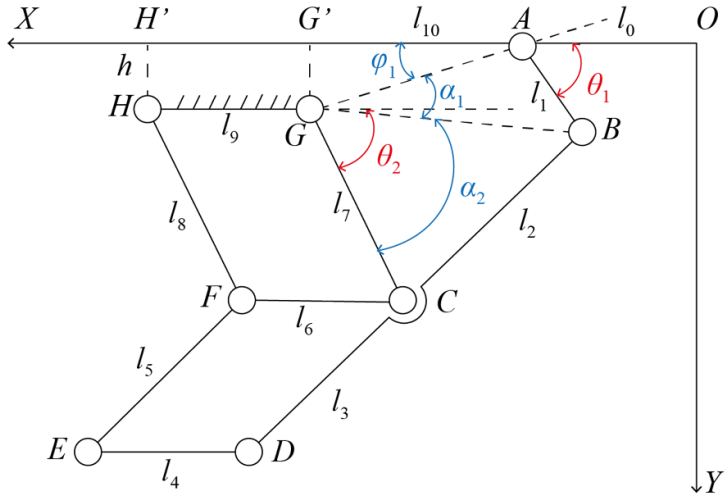

Figure 4. Schematic of the kinematic analysis of the leg mechanism.

During the support phase, the foot remains in contact with the ground, and its trajectory forms a straight line due to relative motion. During the swing phase, the foot lifts off the ground, determining the step length and step height, which significantly influence the motion performance of the leg [34]. This paper adopts a modified cycloid for the swing phase curve and a straight line for the support phase curve. To ensure smooth transitions between the support and swing phases without abrupt changes, a quintic polynomial interpolation is applied.

The planned trajectory refers to the ideal end-effector path designed from kinematic planning, while the actual trajectory represents the simulated motion of the mechanism after optimization, incorporating real linkage dimensions and joint limits. The step length of the robot is $L = 39$ mm, the step height is $H = 9$ mm, and the height from the installation point of the crank to the ground is $H_0 = 50.29$ mm. The trajectory planning for swing phase expression is as follows:

For $(0 \leqslant \varphi \leqslant \phi)$:

$$\begin{cases} S_x = a_1\varphi^5 + b_1\varphi^4 + c_1\varphi^3 + f_1 - \dfrac{L}{2} \\ S_y = H_0 - \left(a_2\varphi^5 + b_2\varphi^4 + c_2\varphi^3\right). \end{cases} \tag{10}$$

For $(\phi \leqslant \varphi \leqslant \pi - \phi)$:

$$\begin{cases} S_x = \dfrac{L}{2\pi}\left(2\varphi - \sin 2\varphi\right) - \dfrac{L}{2} \\ S_y = H_0 - \dfrac{H}{2}\left(1 - \cos 2\varphi\right). \end{cases} \tag{11}$$

For $(\pi - \phi \leqslant \varphi \leqslant \pi)$:

$$\begin{cases} S_x = \dfrac{L}{2} - a_1(\pi - \varphi)^5 - b_1(\pi - \varphi)^4 - c_1(\pi - \varphi)^3 - f_1 \\ S_y = H_0 - \left(a_2(\pi - \varphi)^5 + b_2(\pi - \varphi)^4 + c_2(\pi - \varphi)^3\right). \end{cases} \tag{12}$$

The expression for the support phase in the global coordinate system is as follows:

$$S_y = H_0 \quad \left(\frac{25}{9} - \frac{L}{2} \leqslant S_y \leqslant \frac{L}{2} - \frac{25}{9}\right) \tag{13}$$

where, S_x represent the horizontal coordinate of the trajectory, S_y represents the vertical coordinate of the trajectory, φ represent the phase angle, $\phi = \pi/9$, $a_1 = -3217.51$, $b_1 = 2806.41$, $c_1 = -644.81$, $f_1 = 2.78$, $a_2 = 212.01$, $b_2 = -224.73$, and $c_2 = 77.37$.

To ensure that the leg mechanism follows the planned trajectory, we perform optimization of the leg linkages based on the planned trajectory. Assuming the crank rotates uniformly, the angle rotated by the crank relative to its initial position is denoted as $\Delta\theta$. The planned trajectory is discretized into 13 points, and the crank angle relative to its initial position at each point is $\Delta\theta = 0, \pi/6, \ldots, 2\pi$, with a phase difference of $\pi/6$. Thus, $\theta_1 = \theta_{10} + \Delta\theta$. From the coordinates of point D, it can be deduced that both D_x and D_y are functions of the link lengths and the crank angle. Therefore, the design variables are defined as $\mathbf{x} = [l_1, l_2, l_3, l_4, l_5, l_6, l_7, l_8, l_9, l_{10}, \mathbf{h}, \theta_1]$.

To ensure that the actual trajectory of the foot of the robot is as close as possible to the planned trajectory, an objective function is defined. This function minimizes the root mean square error between the actual trajectory of point D and the given trajectory. The expression for the objective function is:

$$\min f(x) = \sqrt{\frac{1}{13}\sum_{i=1}^{13}\left[\left(D_{xi} - S_{xi}\right)^2 + \left(D_{yi} - S_{yi}\right)^2\right]} \tag{14}$$

where, S_{xi} and S_{yi} are the theoretical coordinates of the ith point of point D, while D_{xi} and D_{yi} are the actual coordinates of point D at the ith position.

Table 5. Optimization of front and rear bar dimensions (Unit: mm).

Rod	l_1	l_2	l_3	l_4	l_5	l_6	l_7	l_8	l_9	l_{10}	h
Before	7.00	23.00	19.00	15.00	19.00	15.00	19.00	19.00	15.00	19.00	5.45
After	9.50	25.00	20.26	15.50	20.26	15.06	20.26	20.26	15.50	20.26	5.02

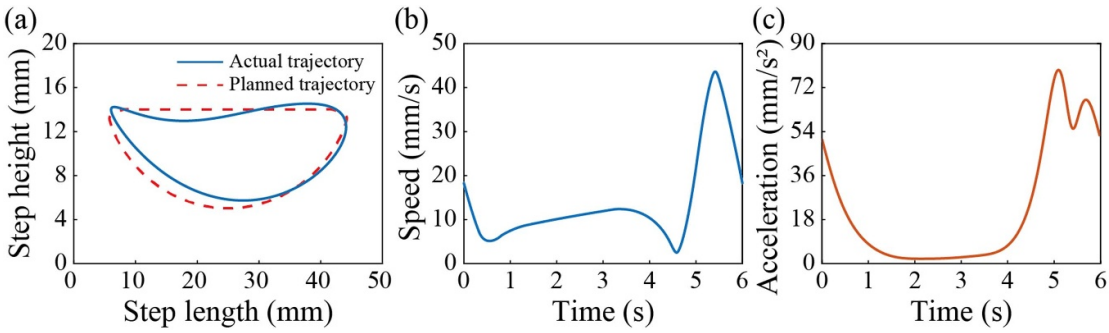

Figure 5. Optimization of the leg mechanism linkages. (a) Comparison between the actual trajectory and the planned trajectory. (b) Velocity curves. (c) Acceleration curves .

In addition, certain constraint conditions need to be specified to ensure that the optimized linkages better meet the design requirements. First, dimensional constraints are applied, defining the allowable length range for each linkage and the initial position of the crank. Moreover, additional constraints arising from the crank-rocker and parallelogram mechanisms are introduced. The overall constraints are as follows:

$$\begin{cases} l_{i_{\min}} \leqslant l_i \leqslant l_{i_{\max}} \quad (i \in \mathbb{Z}^+, 1 \leqslant i \leqslant 10) \\ 0 \leqslant \theta_1 \leqslant 2\pi \\ l_1 + l_2 \leqslant l_7 + l_{AG} \\ l_1 + l_{AG} \leqslant l_7 + l_2 \\ l_1 + l_7 \leqslant l_{AG} + l_2 \\ l_7 = l_8 \\ l_3 = l_5 \\ l_4 = l_6 = l_9 \end{cases} \tag{15}$$

Finally, the multi-parameter nonlinear optimization function *fmincon*(), from the Optimization Toolbox in MATLAB, was used as the optimization algorithm to optimize the lengths of the various linkages [35]. A comparison of the linkage dimensions before and after optimization is shown in table 5.

The optimized leg linkage lengths are incorporated into the leg mechanism, and the simplified mechanism is modeled in ADAMS for basic motion simulation. The simulated actual trajectory is compared with the planned trajectory, as shown in figure 5(a). Although the actual trajectory does not fully overlap with the planned trajectory, the general trends in the stance and swing phases align with the planned trajectory. The maximum deviation between the optimized stance phase trajectory and the planned trajectory is 3 mm, while that of the swing phase is 1.5 mm, resulting in minimal impact on the movement of the robot. Furthermore, the simulation provides the velocity and acceleration curves of a single leg during one gait cycle, as depicted in figures 5(b) and (c). The results indicate that using the planned curve as the foot-end trajectory ensures continuity in the trajectory, velocity, and acceleration during motion without abrupt changes. Moreover, the curve exhibits excellent smoothness, further validating the feasibility of the planned trajectory.

3. Experiments and results

3.1. Prototype development

The robotic mechanical system is mainly divided into three components: the head mechanism, the body mechanism, and the buttock mechanism. The head mechanism consists primarily of a head shell, a servo, a pair of gears, a set of linkages, and a pair of incisors. The body mechanism consists of a body shell, a servo that drives the head, and four sets of leg components. Each set of leg components includes a servo and a set of linkages. The head servo serves to rotate the entire head mechanism, which facilitates soil discharge by the incisors and increases their effective occlusal area. The buttock mechanism is primarily designed to house the control system, with the prototype shown in figure 6(a). Given that the robot is currently in the early research stage, 3D printing technology is adopted to enable rapid prototyping of the robot. Except for the gears, which are made of aluminum alloy, all other printed components are made from resin. The parameters of the robot are presented in table 6, where L, W, and H denote the length, width, and height, respectively.

The control system of the robot is primarily divided into three components: the hardware layer, the driver layer, and the control layer, as shown in figure 6(b). The hardware layer comprises the

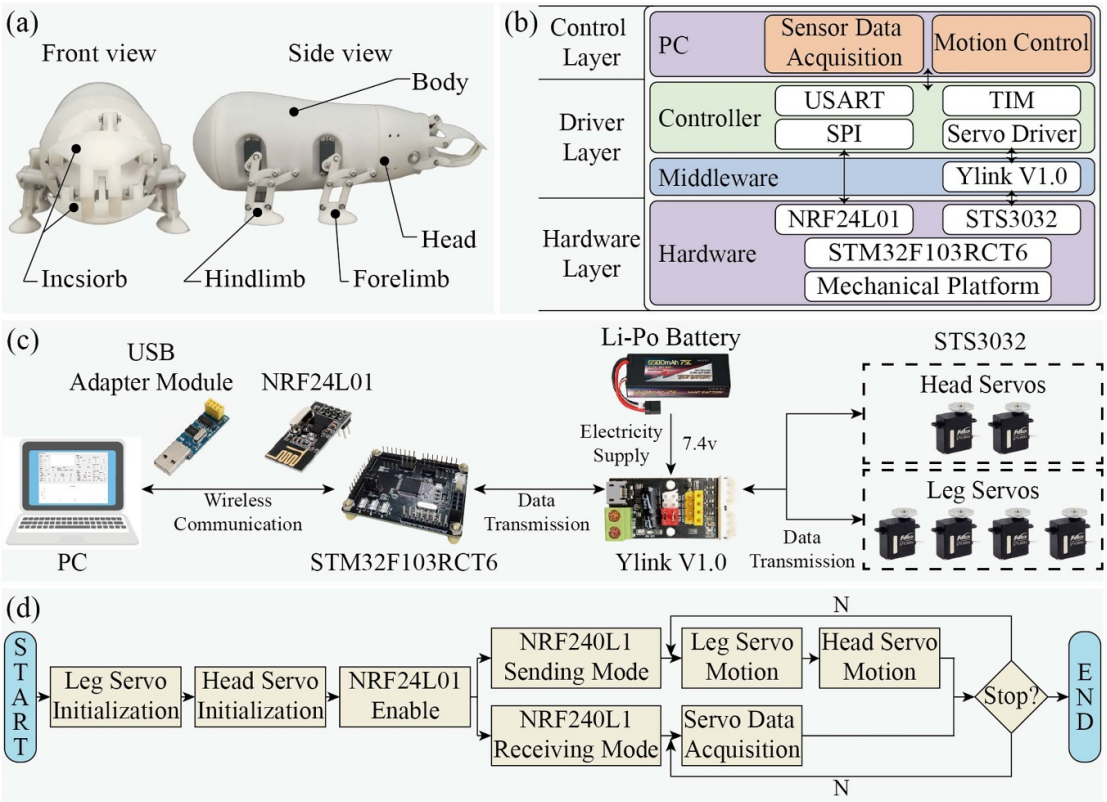

Figure 6. Development of the prototype and design of the control system. (a) Prototype of the robot. (b) Control system of the robot. (c) Hardware components of the control system of the robot. (d) The control process of the robot.

Table 6. Robot physical prototype parameters.

Place	$(L \times W \times H)$/mm	Weight/g	Servo	DOF
Robot	$255 \times 119 \times 128.5$	488	6	6
Head	$99 \times 69 \times 70$	87	2	2
Body	$129 \times 119 \times 128.5$	267	4	4
Buttock	$27 \times 82 \times 82$	134	0	0

mechanical platform (i.e. the mechanical system of the robot), a custom-designed STM32F103RCT6 minimum system board, a wireless communication module (NRF24L01), and servos (STS3032), as shown in figure 6(c). The driver layer consists of a controller and middleware. The controller integrates USART, TIM, SPI, and servo drivers. As the servos cannot directly communicate with the STM32F103RCT6, a servo driver board (Ylink V1.0) is used as middleware. It provides power to the servos and converts standard serial data into half-duplex format, enabling the servos to receive and execute commands. During operation at 12.6 V, the average current per servo ranges from 0.03 to 0.1 A, depending on soil hardness and task load. The control layer incorporates QT software to develop an upper computer tailored for the robot. This upper computer facilitates motion control and enables the collection and storage of feedback data from the servos. The control layer of the robot employs a foreground-background architecture, which is strategically designed through the trade-off analysis between functional requirements and system performance. This architecture has been rigorously validated to fulfill the core functional requirements of the robot, including sensor data acquisition and kinematic motion control. For real-time performance assurance, the system utilizes a software timer interrupt with a 100 ms cycle as the control time base, corresponding to a 50 Hz sampling frequency. The control process of the robot is depicted in figure 6(d).

During the burrowing process, the robot adopts a sequential control strategy consisting of several coordinated actions. Before anchoring, the incisors are first opened, followed by the forward motion of the legs. The incisors then close to bite and detach the soil, after which the legs move backward to withdraw. Finally, the soil held between the incisors is released. This sequence-incisor biting followed by leg propulsion-offers significant advantages when operating in hard or compacted soil environments. The incisors, with their sharp and rigid structure, first penetrate and disrupt the soil integrity, while the subsequent leg propulsion pushes the body forward, enabling efficient excavation and soil removal.

3.2. Occlusal force measurement of the incisors

In section 2, the mechanical model of the incisor mechanism is developed. To validate the accuracy of

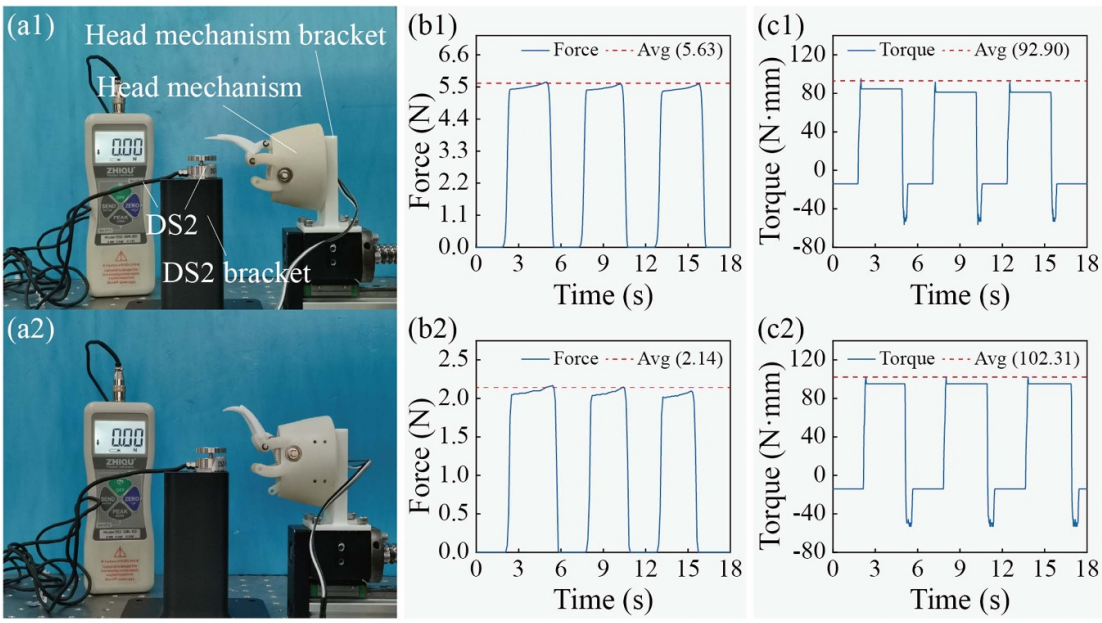

Figure 7. Force measurement experiment for the incisors. (a) Experimental setup for force measurement. (b) Force data obtained from the pressure sensors. (c) Output torque of servo.

this model, force measurement experiments are conducted. The experimental setup utilizes an external micro-sensor digital pressure gauge (DS2) with a maximum load capacity of 50 N and a measurement accuracy of 0.01. This device is equipped with a PC interface, enabling real-time data monitoring and acquisition via a host computer. Since the bottom of the external pressure sensor is unable to measure force, a custom platform is designed to secure the sensor.

In this experiment, independent force measurements were conducted for the upper and lower incisors, with each set of experiments repeated three times. The experimental setup is shown in figures 7(a1) and (a2). Figures 7(b1) and (b2) presents the force measurement data collected by the DS2 via the host computer, where each figure contains three waveforms corresponding to the three experimental repetitions. The maximum value from each experiment was averaged to obtain a single maximum average line (indicated by the red dashed line in the figures). The results indicate that the upper incisor achieves a maximum average biting force of 5.63 N, while the lower incisor achieves 2.14 N. Furthermore, the servo motor driving the incisors provides motion-related data via its host computer. Since the recorded data represent the torque duty cycle (n), conversion using equation (16) is required to derive the actual torque, where K_T is defined as the torque coefficient of the actuator (integrating the peak stall torque and unit conversion factors). For the target motor, K_T is calibrated as 441 N·mm (corresponding to the peak stall torque of 4.5 kg·cm). The converted torque results are presented in figures 7(c1) and (c2). Similarly, the maximum values of the torque data were averaged to obtain a single maximum average line (also indicated by the red dashed line). The results reveal that the maximum output torque of the upper incisor is 92.90 N·mm, while that of the lower incisor is 102.32 N·mm.

$$T_1 = \left(\frac{n}{1000}\right) \cdot K_T \tag{16}$$

The theoretical maximum biting force of the upper and lower incisors is calculated by substituting the output torque of the servo into the previously established mechanical model. Figure 8 presents a comparison between the theoretical values and the measured values. The results indicate a minimal discrepancy, with a maximum error margin of approximately 7.4%. This small error demonstrates the accuracy and reliability of the mechanical model derived in the earlier analysis.

3.3. Burrowing experiment of the robot

This experiment evaluates the burrowing performance of the robot by placing it in sand with varying moisture contents and measuring its burrowing depth and servo output torque. The experimental platform is shown in figure 9. The sand used was initially dry and prone to collapse, necessitating the addition of moisture to enhance bonding force and facilitate the creation of an experimental environment. However, varying moisture levels in the sand result in differing degrees of bonding force, which directly influence the hardness of the sand blocks and, consequently, the burrowing capability of the robot.

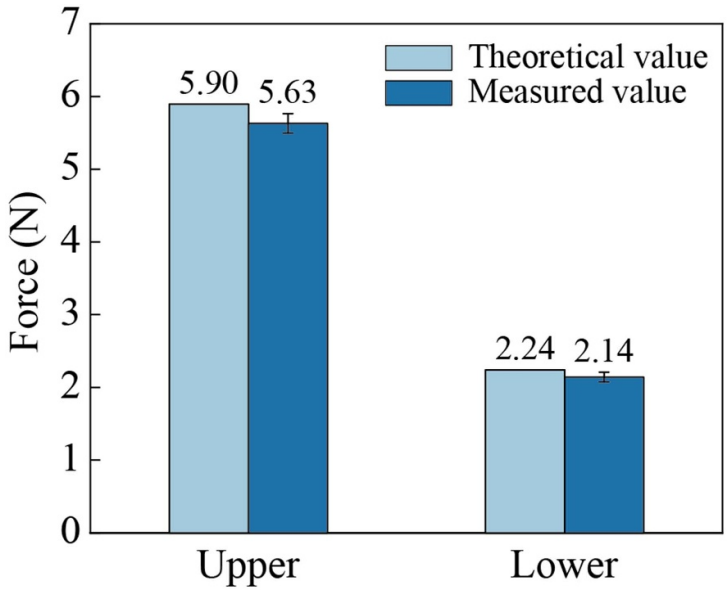

Figure 8. Comparison of theoretical and measured values.

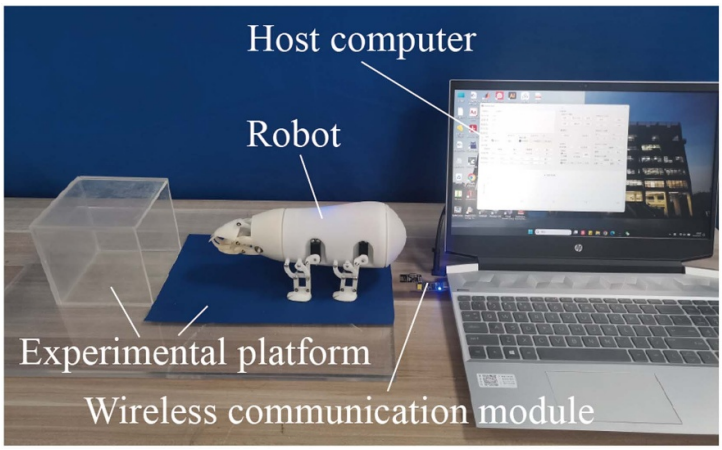

Figure 9. Experimental platform for burrowing experiments.

Based on existing papers, the degree of aridity in various regions can be classified by moisture content: less than 5% represents severe drought, 8%–10% moderate drought, 12%–15% mild drought, 15%–20% wet conditions, and above 20% moist conditions. This experiment involved preparing sand with moisture contents of 5%, 10%, 15%, 20%, and 25%. To ensure consistent initial conditions, the sand was dried using a high-temperature dryer operating at 0.5 kW and 130 °C. The sand was stirred every 10 min to ensure even drying, with a total drying time of 30 min. The moisture content of the sand was determined using the gravimetric method described in equation (17) When the moisture content exceeded 20%, the sand surpassed its water absorption limit and preventing the formation of sand blocks necessary for the experiment. Consequently, only four groups of experiments were conducted.

$$M = \frac{(B - A)}{(B - C)} \times 100\% \tag{17}$$

where M represents the moisture content, A denotes the total mass of the dry sand and container, B represents the total mass of the wet sand and container, and C is the mass of the empty container.

In this experiment, the robot consistently started from the same initial position, with real-time control and data acquisition managed through the host computer. Each experiment was repeated three times. Due to the absence of additional sensors on the robot, the burrowing performance was assessed using the burrowing depth and the output torque of the servo as evaluation metrics.

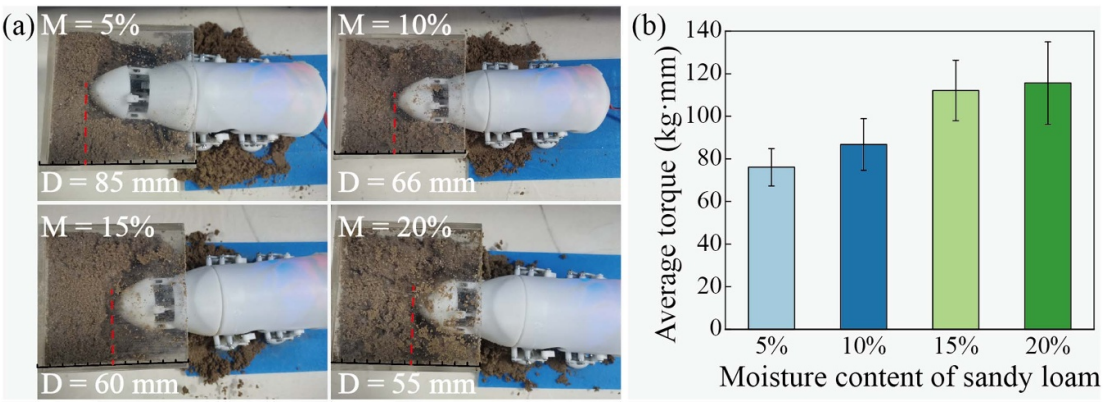

Figure 10. Experiments in varying soil environments. (a) Robot burrowing depth in varying soil environments (M: moisture content; D: burrowing depth). (b) Average torque output of the incisor servo (the rated torque: 147 N·mm).

The burrowing experiment aims to evaluate the burrowing performance of the robot in sand environments with varying moisture contents. As shown in figure 10(a), the robot is capable of burrowing to certain depths in all tested sand conditions. The best performance is achieved in low moisture sand (5%), where the maximum burrowing depth reaches 85 mm. As the moisture content increases, the soil bonding force becomes stronger, leading to higher soil hardness and increased resistance in front of the robot, which limits the achievable burrowing depth. Figure 10(b) presents the average output torque of the incisor servo during burrowing in sand with varying moisture contents. The results indicate that the incisor mechanism can generate sufficient torque to break through soils of varying moisture levels. Moreover, the measured torque values in all conditions remain below the rated torque of the servo. Experimental observations revealed that the limited burrowing depth was primarily caused by soil accumulation beneath the robot body, leading to partial blockage of the passage. The limbs maintained propulsion, and no anchoring failure occurred; however, the clogged passage reduced excavation efficiency. In the current prototype, the limb displacement mechanism has not yet been realized, which contributes to soil accumulation and limited burrowing depth. Future efforts will focus on optimizing the limb design and implementing soil-discharge functionality to achieve deeper and more stable burrowing. This study is currently only a proof of concept, serving as an initial validation of the proposed coupled incisor–limb burrowing mechanism.

Overall, the experimental results validated the burrowing performance of the robot in sand, demonstrating the rationality and practicality of its design. At the same time, they revealed the challenges the robot faces when burrowing in high moisture contents sand (high hardness), providing clear directions for future optimization. Although experiments in this study were conducted using sand to ensure repeatability, the design concept and driving mechanism can be extended to other soil types. Nevertheless, cohesive soils or clay-rich soils may impose greater resistance, necessitating improvements in propulsion and soil discharge capabilities, which will be investigated in future work.

4. Summary and outlook

Inspired by the unique body structure and remarkable burrowing capabilities of naked mole (Heterocephalus glaber), this study developed an incisor-mole-inspired burrowing robot that integrates incisor-driven soil cutting and limb-assisted propulsion. The mechanical and kinematic models were experimentally validated through force and burrowing tests, confirming the rationality of the proposed design. The force measurement experiments verified the accuracy of the incisor mechanical model, with a maximum deviation of 7.4% between theoretical and experimental results. In burrowing experiments, the robot achieved a maximum burrowing depth of 85 mm in sand environments with varying moisture contents, while the measured servo torque remained below the rated value in all cases, indicating safe and stable operation. These results demonstrate that the proposed incisor–limb coordination mechanism enables efficient soil penetration and propulsion under different soil conditions. This study established a complete closed-loop process from design to modeling and experimental validation, demonstrating that the proposed incisor–limb coordination mechanism achieves effective soil penetration and propulsion under various conditions. Moreover, the proposed design and control methodology can be readily extended to other burrowing or tunneling robots requiring efficient soil fragmentation and propulsion in complex terrestrial or extraterrestrial environments.

Future work will focus on optimizing the leg mechanism design by developing a multidegree-of-freedom structure to enhance leg flexibility, specifically introducing additional degrees of freedom and redesigning the leg structure to improve locomotion flexibility and terrain adaptability, as well as improving soil discharge efficiency to prevent soil accumulation from impeding the movement of robot. The inclusion of cameras and other various sensors is anticipated to facilitate the acquisition of underground environmental information, enable path planning, and be integrated into the system to achieve adaptive, phase-synchronized coordination between the cutting and leg modules, allowing the robot to dynamically adjust its motion sequence in real time according to variations in soil resistance and thereby enhancing its adaptability to complex and changing environments. Meanwhile, more advanced control algorithms will be implemented to enable multiple motion modes, and leveraging the multi-DOF leg mechanism, the robot will be capable of performing active soil-clearing motions to effectively prevent soil accumulation beneath the chassis. Finally, we aim to expand the experimental conditions by introducing soil samples with different compaction levels and particle sizes in subsequent studies to systematically quantify the relationship between excavation force and energy consumption, further refine the dynamic model to improve prediction accuracy and control precision, and conduct multiple experimental validations to verify improvements in energy absorption and operational efficiency, thus further validating the adaptability and practicality of the robot.

Data availability statement

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Conflicts of interest

The authors declare that there are no conflicts of interest regarding the funding sources or the conduct of this research.

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